

Foundation Model Insights and a Multi-Model Approach for Superior Fine-Grained One-shot Subset Selection



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Highlights

- ✓ **First comprehensive analysis** of Foundation Models (FMs) vs traditional Information Extractors (IEs) for one-shot subset selection;
- ✓ **FM-based subset selection excel on fine-grained datasets;**
- ✓ **FM-based subset selection offer limited gains on noisy, coarse-grained datasets;**
- ✓ **RAM-APL:** A novel multi-FM selection strategy combining intra/inter-class distributions;
- ✓ **State-of-the-art results** on Oxford-IIIT Pet, Food-101, CUB-200-2011.

Background & Motivation

? Why subset selection?

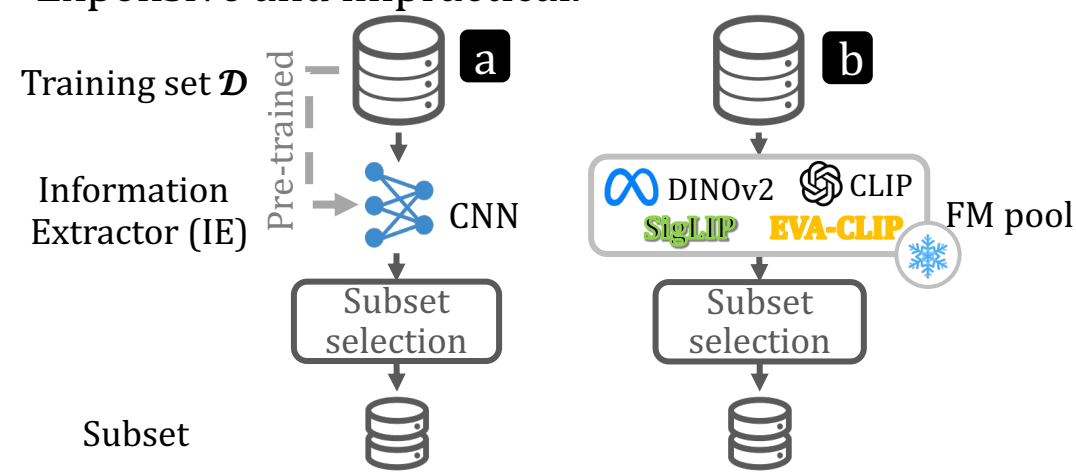
- Data explosion, but more data $\neq$ better learning;
- Training on full data is expensive, often redundant.

Why One-shot?

- Single-pass selection before training;
- Highly efficient and scalable.

Challenge with traditional information extractors:

- Dataset-dependent;
- Training set updated $\rightarrow$ pretrain again;
- Expensive and impractical.



Research Questions

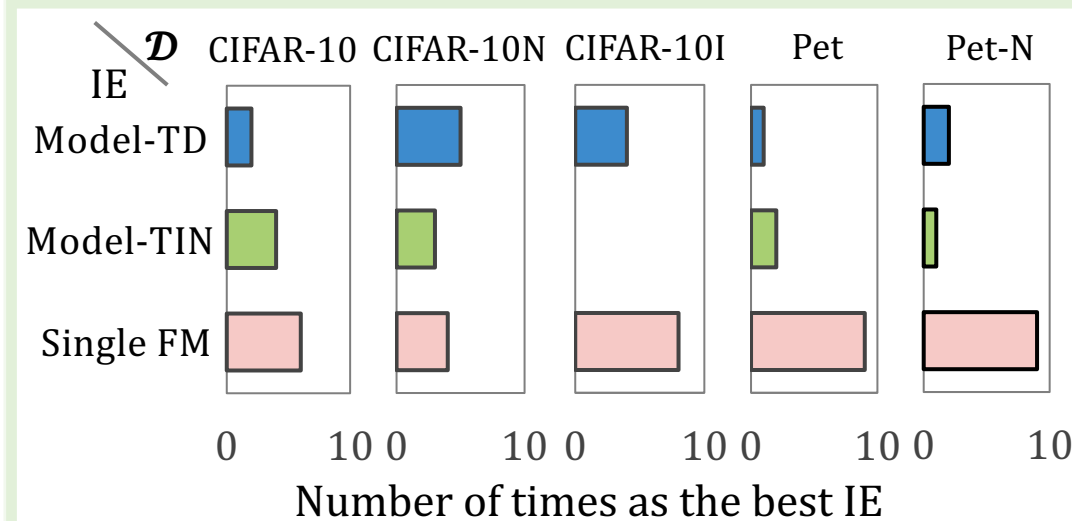


RQ1: Can Foundation Models (FMs) outperform traditional Information Extractors (IEs) for one-shot subset selection across datasets?



Observation 1: FMs demonstrate limited advantages for subset selection on noisy, coarse-grained datasets.

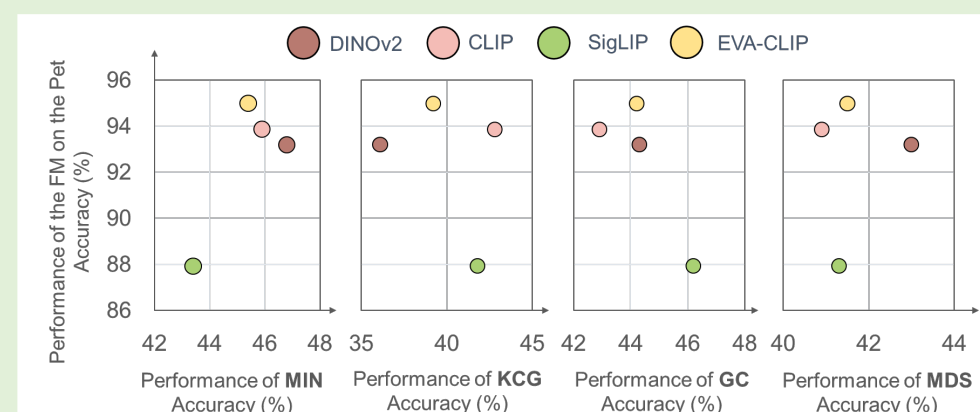
Observation 2: FMs significantly and consistently outperform traditional IEs for subset selection on fine-grained datasets.



RQ2: Are all FMs equally effective?



Observation 3: Different FMs vary in subset selection performance, and stronger downstream accuracy doesn't imply better selection quality.



RAM-APL: Multi-FM Subset Selection

Key idea:

- Leverage complementary strengths of multiple FMs for better fine-grained selection;
- Comprehensive analysis of intra- and inter-class feature distributions:

□ **RAM:** Intra-class ranking mean $\rightarrow$ representativeness

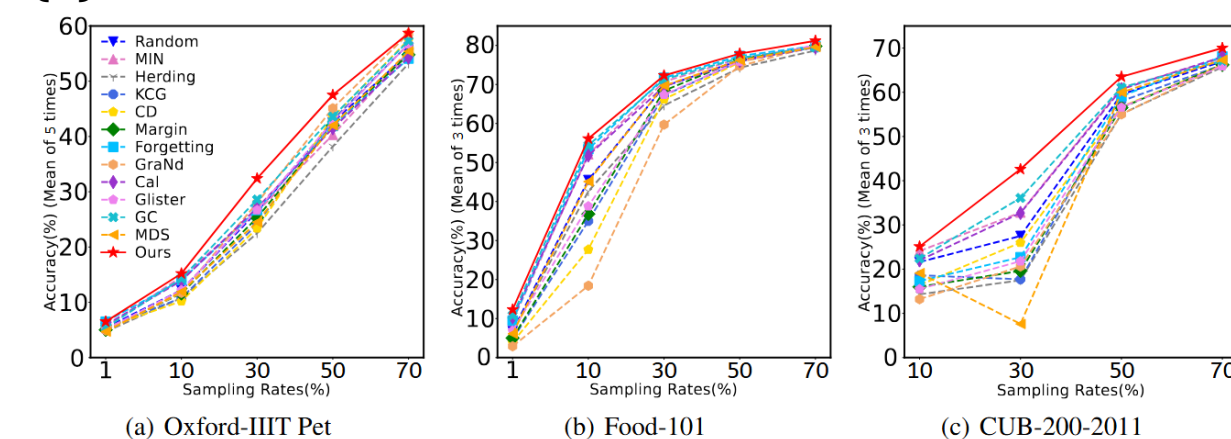
□ **APL:** Pseudo-label accuracy $\rightarrow$ inter-class difficulty

□ **Scoring function** (select those with the lowest scores):

$$Score(x) = W_1 * RAM(x) + W_2 * (1 - APL(x))$$

Experimental Results

(1) State-of-the-art Subset Selection



(2) Multi-FM selection can outperform all single-FM baselines.

Table. Comparison of the performance of our method using different numbers of foundation models as information extractors. Here, "D", "C", "S" and "E" represent DINOv2, CLIP, SigLIP, EVA-CLIP, respectively.

IE	Sampling rates					Overall Mean
	D	C	S	E		
• • • • •	5.9±0.3	15.4±1.1	31.6±2.3	47.7±1.1	57.9±4.1	158.5
• • • • •	5.7±0.4	15.0±0.2	27.9±1.2	43.6±1.9	57.0±0.4	149.2
• • • • •	6.6±0.3	14.1±1.0	28.8±1.1	43.9±1.7	55.1±2.6	148.5
• • • • •	5.4±0.3	15.0±0.6	30.2±2.5	44.4±2.3	56.6±1.8	151.6
• • • • •	6.5±0.4	15.2±1.2	32.4±2.9	47.5±1.9	58.7±2.2	160.3
• • • • •	5.9±0.3	16.2±0.1	31.4±3.2	45.0±1.3	58.6±1.2	157.1
• • • • •	6.0±0.6	16.0±0.9	35.8±2.9	46.5±1.8	54.9±3.5	159.3
• • • • •	6.4±0.2	15.1±0.4	29.8±1.6	45.9±1.3	56.2±2.7	153.4
• • • • •	5.9±0.3	15.5±0.7	31.4±1.7	44.2±2.2	55.9±1.8	152.9
• • • • •	6.7±0.4	16.2±0.6	34.7±0.3	45.7±0.8	56.6±2.4	159.9
• • • • •	6.2±0.8	15.6±0.5	33.2±1.4	48.3±1.1	57.6±0.1	160.9
• • • • •	6.0±0.4	17.5±1.0	35.2±1.8	47.9±1.5	55.6±2.1	162.2
• • • • •	6.1±0.3	16.8±0.6	34.4±2.1	47.0±2.0	55.1±1.6	159.4
• • • • •	6.1±0.2	16.1±0.3	33.9±1.4	46.8±1.5	55.1±0.5	158.0
• • • • •	6.5±0.2	16.8±1.1	34.0±2.7	46.3±0.5	56.9±1.1	160.5

Best efficiency | accuracy trade-off

Paper



Code

